Chapter 10 - Pair of Straight Lines

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#General Equation of Second Degree

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

#Homogeneous Equation of Second Degree

An equation with two variables x and y in which each term has the same degree is known as a homogeneous equation. If the degree of each term is 2 then it is called a homogeneous equation of second degree.

General form of a 2nd degree homogeneous equation: $ax^2 + 2hxy + by^2 = 0$

#Angle between the line pair represented by $ax^2 + 2hxy + by^2 = 0$

If θ is the angle between the pair of lines represented by $ax^2 + 2hxy + by^2 = 0$ then

$$\tan\theta = \pm\frac{2\sqrt{h^2 - ab}}{a+b}$$

Condition of perpendicularity: $a + b = 0$

Condition for coincident lines $h^2 = ab$

#Bisectors of the Angles between the Pair of Lines represented by $ax^2 + 2hxy + by^2 = 0$

If a pair of straight lines is represented by $ax^2 + 2hxy + by^2 = 0$ then equation of the bisectors of angles between the given lines is $h(x^2 - y^2) = (a - b)xy$.

#Condition that the General Equation of Second Degree may Represent a Line Pair

The equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if

$abc + 2fgh = af^2 + bg^2 + ch^2$

#Lines joining the Origin to the Intersection of a Line and a Curve

The equation of pair of lines passing through origin to the intersection of a line $lx + my = n$ and a curve

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is

$$ax^2 + 2hxy + by^2 + 2gx\left(\frac{lx+my}{n}\right) + 2fy\left(\frac{lx+my}{n}\right) + c\left(\frac{lx+my}{n}\right)^2 = 0$$